

Some Effects of Composted Organic Materials on Plant Health*

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ABSTRACT

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Watery fermentation extracts of well-composted organic materials reduced disease incidence and intensity in various host–pathogen combinations, if applied prophylactically to plant surfaces. Successes were achieved with the following systems: *Plasmopara viticola*, *Uncinula necator* and *Pseudopeziza tracheiphila* on grape vine; *Phytophthora infestans* on potatoes and tomatoes; *Erysiphe graminis* on barley; *Erysiphe betae* on sugar beets; *Sphaerotheca fuliginea* on cucumbers and *Botrytis cinerea* on strawberries and beans. Induced resistance is one of the mechanisms involved, but direct fungus inhibition has been observed.

INTRODUCTION

Biological control of fungal plant diseases has not yet been achieved to any large extent under outdoor conditions. Schönbeck et al. (1980) have reported induced resistance after the application of soil microorganisms and their culture filtrates against various obligate parasites. Treatment of planting material with suspensions of fluorescent *Pseudomonas* spp. from field soils did suppress pathogenic organisms such as *Erwinia carotovora* (Rhodes and Logan, 1986). Composted organic soil amendments have been used successfully to suppress soil-borne pathogens (Hoitink, 1980; Lumsden et al., 1983; Hoitink and Fahy, 1986; Hadar and Mandelbaum, 1986). Therefore we have studied the effects and possible applications of composted organic materials on plant health, using various host–parasite systems.

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MATERIALS AND DEFINITIONS

Composts were understood to be largely degraded organic materials with the original organic structures no longer recognizable. They were composed of animal manure (horse, hog, goat) with cereal straw and amendments of top soil and/or old compost of the same origin. Composts were used after 8–12 months. For soil amendments, composts were mixed with ordinary field soil.

Watery extracts of composts were obtained by covering compost with tap water at a ratio of 1:5–1:8. They were stirred only once and then left for extraction and fermentation under laboratory or outdoor conditions, with temperatures usually ranging between 15 and 20°C. The period, during which the compost was covered with water, was called “extraction time”. For application, the broth was filtered through cheese cloth. If sprayed, ordinary laboratory, greenhouse and field sprayers were used. The time span between application and inoculation of a pathogen was called “induction time”. Details of the host-parasite systems used and the specific experimental conditions are given below.

RESULTS

*Diseases of grape vines (Vitis vinifera)**Downy mildew, Plasmopara viticola*

Fully grown leaves from greenhouse plants (cv. ‘Müller-Thurgau’) were placed in petri dishes, treated with compost extract and later inoculated with sporangia suspensions of $7-9 \times 10^4$ sporangia ml⁻¹. Incubation followed at 18–20°C with 16 h light/day of 1700 lux (Weltzien and Ketterer, 1986). Under outdoor conditions we used a vineyard in the Ahr valley near Bonn. As no artificial inoculation could be used there, and spontaneous leaf infections did not occur in 1986, we were limited to the evaluation of infected berries (Ketterer and Weltzien, 1987). Some of the results are summarized in Table 1. Horse-manure compost was used throughout.

The results clearly indicate, that downy mildew can be largely suppressed by prophylactic compost extract treatment. Further laboratory and outdoor experiments are being performed to confirm and improve the results obtained so far.

Powdery mildew (Uncinula necator) and leaf blight (Pseudopeziza tracheiphila)

Results with both host-pathogen systems were achieved under greenhouse and outdoor conditions, using again the ‘Müller-Thurgau’ cultivar. Compost extracts from horse-manure compost were compared with untreated checks and commercial fungicides. Some results are summarized in Table 2 (Ketterer and Weltzien, 1987).

TABLE 1

Effect of compost extract application on grape vine (cv. 'Müller-Thurgau') against *Plasmopara viticola* attack on leaves and berries

Extraction time (days)	Application	Induction time	Leaf area covered × of 10 leaves		No. of berries × of 500 berries	
			Treated	Check	Treated	Check
1	Leaves dipped					
	3 h	3 h	92	86	—	—
3	Leaves dipped					
	3 h	3 h	2.2*	83	—	—
3	Leaves sprayed	3h	1.6*	90	—	—
3	Plants sprayed weekly	Spontaneous infection	—	—	3.2*	10.5*

*Different from check at < 0.1% statistical significance.

TABLE 2

Effect of compost extract application on grape vine (cv. 'Müller-Thurgau') against *Uncinula necator* (powdery mildew) and *Pseudopeziza tracheiphila* (leaf blight)

Treatment	Disease incidence in % of check	Pathogen
Compost extract, 3 days extraction time every 2 weeks, greenhouse	8	<i>U. necator</i>
Wettable sulfur every 2 weeks, greenhouse	12	<i>U. necator</i>
Compost and wettable sulphur every 2 weeks, greenhouse	7	<i>U. necator</i>
Compost extract, 3 days extraction time weekly, outdoors	50	<i>P. tracheiphila</i>
Fungicides, as recommended outdoors	32	<i>P. tracheiphila</i>

All differences significant against check at 5% level.

The effect against powdery mildew under heavy infection pressure in the greenhouse was remarkable and compared favorably with sulphur treatment. Also the reduction in the incidence of leafblight deserves attention, as it was achieved under outdoor conditions. Here, however, the recommended fungicide treatments resulted in significantly better disease control.

Phytophthora infestans on potatoes and tomatoes

The results with the oomycete *P. viticola* encouraged us to extend our studies to *Phytophthora infestans* on potatoes (cultivar 'Grata') and tomatoes (culti-

TABLE 3

Effect of compost extract application on single leaves of potatoes (cultivar 'Grata') and tomatoes (cultivar 'Rheinglut') with 3 days incubation time against *Phytophthora infestans*

Extracted compost	Extraction time (days)	Host plant	Percent leaf area affected
Check	—	Potato	78
Horse manure	7	Potato	8
Hog manure	7	Potato	21
Goat manure	7	Potato	7
Check	—	Tomato	63
Horse manure	7	Tomato	30
Horse manure	14	Tomato	24
Hog manure	7	Tomato	30
Hog manure	14	Tomato	15
Goat manure	7	Tomato	24
Goat manure	14	Tomato	19

All differences significant against check at 5% level.

var 'Rheinglut'). Three different composts were compared with 2 different extraction times and 3 days incubation time. Single leaves were treated, inoculated and incubated under laboratory conditions. Table 3 summarizes some results (Weltzien et al., 1987).

The results indicate the strong effects of compost extracts on *Phytophthora infestans* on both hosts. There were no significant differences between composts based on different organic materials. Both characteristics remained unchanged, if the extraction time was extended from 7 to 14 days.

Powdery mildew of barley, sugar beet and cucumber

Powdery mildew of barley (*Erysiphe graminis* f. sp. *hordei*) is a good model for Erysiphaceae. We experimented with seedlings of winter barley (cultivar 'Gerbel') using primary leaves grown under greenhouse conditions. They were used as cut leaves, placed on benzimidazol-agar, or in situ and inoculated with freshly harvested conidia in an inoculation tower. Results depended to some extent on the compost used, the extraction and the induction time (Weltzien et al., 1987; Budde and Weltzien, 1988). Table 4 summarizes some results.

Though all experiments were significantly different from the checks, they were not significantly different from each other. The average reduction in number of pustules per primary leaf was 55%, indicating a limited but reproducible effect.

To study the effect of compost as a planting substrate on plant health, barley seeds were planted in substrates composed of different field soil-compost mixtures. Then the primary leaves were tested for their resistance as described

TABLE 4

Effect of compost extract application on primary leaves of winter barley (cv. 'Gerbel') against powdery mildew, *Erysiphe graminis* f. sp. *hordei*

Extracted compost	Extraction time (days)	Induction time (days)	No. of pustules per primary leaf in % of check
Check, dry	—	—	100
Check, H ₂ O	—	3	80
Horse manure	3	1	60
Horse manure	3	2	20
Horse manure	3	3	40
Horse manure	7	3	40
Horse manure	21	3	47
Earthworm (commercial)	3	3	43
Cattle manure (commercial)	3	3	60
Goat manure + green plants	3	3	47
Average of all treatments			45

All differences significant against check at 5% level.

TABLE 5

Increased resistance of primary leaves of winter barley (cv. 'Gerbel') to *Erysiphe graminis* through increasing horse manure compost content in the planting substrate and additional compost extract treatment

Planting substrate soil/compost mixture	Extract treatment	No. of pustules per primary leaf as % of check
1:0	—	100
5:1	—	71
2:1	—	74
1:1	—	54
1:2	—	43
1:5	—	21
0:1	—	16
1:1	Extraction time 3 days; Induction time 2 days	7
0:1	Extraction time 3 days; Induction time 2 days	2

All differences significant against check at 5% level.

earlier. In some cases, additional compost extract treatments were applied. Some results are presented in Table 5 (Budde and Weltzien, 1988).

It is apparent, that the increasing compost content of the planting substrates correlates with decreasing disease intensity. If extracts are cumulatively applied, the disease intensity is further decreased.

A more detailed study was performed with powdery mildew of sugar beet

TABLE 6

The effect of compost extract sprays (horse-manure compost) on *Botrytis cinerea* infections of bean leaves (cultivar 'Processor') and strawberry fruits (cultivar 'Corona')

Extraction time	Induction time	Host	<i>Botrytis</i> infection % of check
4 days	2 h	<i>Phaseolus</i> bean leaves	32
7 days	—	Strawberries, field experiment	84
12 weeks	—	Strawberries 1	55
12 weeks	—	Strawberries 2	47
12 weeks	—	Strawberries 3	52

All differences significant against their respective checks at 5% level.

(*Erysiphe betae*) and of cucumber (*Sphaerotheca fuliginea*), to identify the stages of fungus development most heavily affected by compost extracts. Conidia germination was always equal to the check, but the formation of secondary hyphae was reduced by more than 50% (Samerski and Weltzien, 1988a,b).

Botrytis cinerea on beans and strawberries

Botrytis cinerea infection can be tested on detached *Phaseolus* bean leaves. Spore suspensions are applied to the leaflets which have been sprayed with compost extracts 2 days earlier. They were then kept in a moist chamber at 22°C, for 6 days at a 16-h light regime with 8000 lux. Strawberries (cultivar 'Corona') were treated 7 times in weekly intervals between 24 April and 6 June, under field conditions in 3 different locations. Some results are summarized in Table 6 (Stindt and Weltzien, 1988a,b).

The data indicate the potential for *Botrytis* suppression through compost extract application. This is true for both leaf and fruit infections.

DISCUSSION

Compost extract proved to be effective against various fungal diseases of leaves and fruits, if applied prophylactically. The effects were significant and reproducible under varying conditions in the laboratory, the greenhouse, and outdoors. Information on the mechanisms involved is still scarce. It has been shown, that after sterile filtration as well as heat disinfection of the compost extract the broth remains rather ineffective. It seems, that the presence of microorganisms, especially bacteria, is a prerequisite for its biological activity (Stindt and Weltzien, 1988b). If leaves are rinsed 2 days after treatment and before inoculation with powdery mildew spores, the inhibitory effect remains. This shows, that induced resistance is at least partly responsible for the effect in this host-pathogen system (Samerski and Weltzien, 1988a). No systemic

effects could be observed in the case of barley powdery mildew on leaves (Budde and Weltzien, 1988b). However, if barley was planted in compost-enriched soil, the primary leaves acquired some resistance against *Erysiphe graminis* without having been sprayed with extracts before. Further clarification is required to determine if this effect is due to uptake and transport of inhibitory substances or their inductors. If plants grown in compost are treated additionally with compost extract, the effect is cumulative. This shows that the mechanisms involved are probably of a different nature.

Botrytis cinerea infections were also reduced. Additional studies revealed, that this is largely due to direct inhibition of the fungus development (Stindt and Weltzien, 1988a). We may therefore assume that the mechanisms of action of compost extracts differ, depending on the host-pathogen system used.

Successful outdoor experiments show, that a satisfactory control of plant leaf diseases can also be achieved under field conditions. This was especially true, if extracts were enriched with inhibitory active microorganisms isolated from compost (Ketterer and Weltzien, 1988). In summary, the biological control of fungal leaf and fruit diseases may well develop into an economic form of disease control. Much research work remains to be done, but our outlook is optimistic.

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